Planned work and completion checks

The remaining work needed to turn the tested desktop update into a complete local runtime manager and router.

> **Priority change:** Performance validation is now before Additional runtimes. A planned feature is not a promise that it is already running or shipping.

To-do order

Priority	Workstream	Current state
1	**GPU memory planning and reservations**	Not implemented
2	**Independent workers and request routing**	Not implemented
3	**Live GPU and worker monitoring**	Partial: identity and timing utilities exist
4	**Validated runtime updates and rollback**	Not implemented: package inspection exists
5	**Performance validation**	Partial: short correctness canaries passed
6	**Additional runtimes**	Partial: vLLM/TensorRT-LLM management controls exist
7	**Final installer and fresh-install experience**	Partial: portable build/update and source setup exist
8	**Independent review, integration and release**	Pending

How to use this printout

Each workstream has a goal, the remaining implementation work and empty completion-check boxes. Mark a check only after its evidence exists. Dependency notes explain what must be available for the result to be credible.

RAG remains deferred. The roadmap does not assign delivery dates, claim a production-ready percentage or treat a portable ZIP as a finished fresh-install experience.

Source: draft PR #5 and its PROJECT_STATUS.md; source commit 973b2e3. Snapshot checked November 6, 2026.

P01 GPU memory planning and reservations

Not implemented

Goal. Prevent the wrapper's managed workloads from promising the same free VRAM to more than one worker.

Work to complete

- Calculate usable memory after configurable headroom, pending reservations and expected growth.
- Treat stale or missing telemetry as uncertain; avoid double-counting memory already resident on a card.
- Acquire reservations atomically across selected GPUs and release them after failure, cancellation or process exit.

Completion checks

☐ Two simultaneous starts cannot claim the same remaining budget.

☐ Startup failure and out-of-memory paths release their reservations.

☐ Profiles keep their requested placement and budgets; no silent fallback rewrites them.

Dependencies and limits. Reliable inventory and sufficiently fresh memory readings are required. External GPU applications can still consume VRAM outside the wrapper's control.

Reviewed: ☐ Date: _____ Owner: _____

P02 Independent workers and request routing

Not implemented

Goal. Support one large model across cards, a different model on each card, and multiple models sharing a card when memory allows.

Work to complete

- Give each managed model instance its own ID, configuration, process/endpoint and health state.
- Track ownership, free ports, startup/stop behavior and request destinations in a worker registry.
- Add queueing, cancellation and failure cleanup; preserve the lifecycle of attached external providers.

Completion checks

☐ A dual-card model and two per-card models have distinct resolved plans.

☐ Requests reach the intended model and worker; port collisions and cancellation have tested behavior.

☐ Stopping or failing one worker does not incorrectly stop an unrelated provider.

Dependencies and limits. Memory admission from P01 must govern shared-card starts. The current whole-server placement controls do not provide this behavior by themselves.

Reviewed: ☐ Date: _____ Owner: _____

P03 Live GPU and worker monitoring

Partial: identity and timing utilities exist

Goal. Show what each GPU and worker is doing in the existing app controls.

Work to complete

- Display available VRAM, utilization, power and temperature with worker membership.
- Add active requests, queue/load state, tokens per second and client-observed TTFT.
- Refresh at a bounded rate, reduce work when hidden and distinguish missing readings from real zeros.

Completion checks

☐ Each reading identifies its GPU or worker and whether it is current or unavailable.

☐ TTFT and throughput labels use documented definitions and are not confused with connection readiness or server-only timing.

☐ Monitoring survives missing NVML data without invented values or an unresponsive UI.

Dependencies and limits. Worker ownership comes from P02. NVML initialization currently fails on the validation host; CUDA identity discovery does not replace power/utilization telemetry.

Reviewed: ☐ Date: _____ Owner: _____

P04 Validated runtime updates and rollback

Not implemented: package inspection exists

Goal. Obtain compatible runtime updates without losing saved profiles or replacing a working package prematurely.

Work to complete

- Download authoritative artifacts into immutable staged package directories with hashes and version records.
- Check required options and run isolated load/generation canaries for the intended model/build.
- Wait for affected workers to drain, journal activation and retain known-good packages for recovery.

Completion checks

☐ Corrupt packages, missing flags and failed canaries leave the active runtime intact.

☐ Locked DLLs, interrupted activation and failed restarts have tested recovery paths.

☐ Profiles survive update and rollback; the active package is identifiable.

Dependencies and limits. This depends on capability inspection and worker ownership. The existing pinned b12125 package and fingerprints are a foundation, not an automatic updater.

Reviewed: ☐ Date: _____ Owner: _____

P05 Performance validation

Partial: short correctness canaries passed

Goal. Establish repeatable performance and capacity evidence before expanding runtime coverage.

Work to complete

- Use fixed model hashes, quantization, prompts, generation limits, context and cache policies.
- Compare one 3090 with dual-3090 layer mode; test tensor mode only when that exact model/build is eligible. Add mixed-card tests only when hardware and memory permit.
- Separate cold loading from warm inference, warm up, then repeat runs at concurrency 1, 2 and 4 where admission allows.

Completion checks

☐ Retain raw results plus runtime, adapter, model, driver and topology identities.

☐ Report median/tail latency, TTFT, actual output-token counts, throughput, errors and peak VRAM; include power/energy only when reliable sampling is available.

☐ Check correctness before accepting a faster configuration and avoid general claims from a single small-model run.

Dependencies and limits. This work is explicitly prioritized before P06 Additional runtimes. Monitoring and reservations improve the evidence; each comparison must state any measurements still unavailable.

Reviewed: ☐ Date: _____ Owner: _____

P06 Additional runtimes

Partial: vLLM/TensorRT-LLM management controls exist

Goal. Expand the router through runtime-specific adapters after the performance-validation priority.

Work to complete

- Finish real model/endpoint validation for the existing WSL vLLM and TensorRT-LLM controls.
- Add KoboldCpp, Ollama and LM Studio integration using the interfaces each runtime actually supports.
- Provide isolated, versioned worker environments for Transformers; retain ExecuTorch as a target requiring artifact/delegate validation. Cloud-provider routing is a later requested direction.

Completion checks

☐ Each adapter states which controls are supported, unsupported or unknown for its detected version.

☐ Health, model identity, requests and lifecycle behavior are verified without pretending llama.cpp flags are universal.

☐ Do not claim generic Windows multi-GPU serving for an untested runtime or delegate.

Dependencies and limits. P05 Performance validation comes first in the roadmap. Runtime installation, model compatibility and management authority must be established separately for each adapter.

Reviewed: ☐ Date: _____ Owner: _____

P07 Final installer and fresh-install experience

Partial: portable build/update and source setup exist

Goal. Let a user install and launch the completed app with clear prerequisites, a working shortcut and no manual server-start sequence.

Work to complete

- Finish the installer flow, prerequisite checks, installation paths and first-run setup.
- Preserve model locations, saved profiles, caches and logs during application updates.
- Verify upgrade, uninstall/reinstall and fresh-machine behavior, including informative errors when a dependency is missing.

Completion checks

☐ A clean supported Windows installation can start the intended server and UI through the installed shortcut.

☐ Existing model files and settings survive an update.

☐ The installer is tested as an installer; a portable ZIP alone does not satisfy this gate.

Dependencies and limits. The final installer must package the intended completed behavior. The current update is for an existing portable installation.

Reviewed: ☐ Date: _____ Owner: _____

P08 Independent review, integration and release

Pending

Goal. Turn the tested development update into a reviewed, integrated release with an accurate support boundary.

Work to complete

- Complete independent review, resolve findings and run the relevant CI/end-to-end checks on the final commit.
- Review the draft PR, integrate approved changes and build versioned release artifacts.
- Publish release notes, validation evidence and known limitations that distinguish implemented controls from verified serving support.

Completion checks

☐ The reviewed code and release artifacts identify the same source state.

☐ Required checks and reviewer findings are resolved or explicitly documented.

☐ Published instructions and the landing page match what a new user actually receives.

Dependencies and limits. PR #5 is open and draft as of this snapshot. No production-ready completion percentage or delivery date is asserted.

Reviewed: ☐ Date: _____ Owner: _____